EPSR-test-ex54 torangles: torsion-angle distributions

Program(s): torangles | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `torangles`, which computes intramolecular torsion (dihedral) angle distributions [1]: for every molecule whose first atom has type `atom1`, the dihedral about the axis between molecule atoms `ltyp1`-`ltyp2`, the end points being the mid-points of the atom pairs `ltyp3` (bonded to the first axis atom) and `ltyp4` (to the second). Distributions over -180..180° are accumulated and each molecule's angles are listed.

Code exercised

Fortran (original, fortran_src/):

- `torangles.f`: `run_torangles`, `find_torangles` (dihedral from mid-points, angle list)
- `auxiliary_routines.f`: `vecpro`, `scapro` (vector and scalar products)
- `read_and_save_files.f`: `read_input_file`, `initgofz`, `outgofz` (histogram file, running average and spread), `save_files_others`; `auxiliary_routines.f`: `r_int`
- `setup_symbols_generic.f`: `setup_symbols_torangles`

Rust (port, rust/epsr-core/src/):

- `tools/torangles.rs`: `run`, `find_torangles`
- `tools/anal_common.rs`: `outgofz`; `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `TORANGLES`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å³), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [2]. Torsions (atom numbers within the molecule: DMSO S1, C2, C3, O4, H5-H10): O4-S1-C2-H5, the methyl rotation relative to the S=O bond; C2-S1-C3-mid(H8,H9), using the mid-point of two methyl hydrogens; and a degenerate water 'torsion' O-H-H with all end points on atom 3 (`ltyp3 3 3`, `ltyp4 3 3`). 72 bins of 5°. In the EPSR model the DMSO methyl groups are rotational groups (`ROT` entries in the `.ato` file), so the H5 torsion samples the methyl rotation.

Inputs

`dmso.TOR.dat`: `nsize 72`, `nsumt 0`, three distributions as above (`atom1 S` twice, `atom1 O1`); `200DMSO400Water.ato` copied unchanged from EPSR-test.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`repeat 2` with `nsumt 0`: pass 2 reads `dmso.TOR.c01` back and averages. Compared after each pass: `dmso.TOR.c01` (three distributions with spreads), the rewritten `dmso.TOR.dat` and `toranglesvalues.txt`, the angles of every molecule written with the run-time format `(1x,i6,n(1x,e14.7))` built by the program. The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.